

Physics 101: Mechanics

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Contents

Chapter 1

The Distance Formula

Let's analyze the distance formula, which is an application of the Pythagorean theorem. The formula is given by the equation:

distance formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (1.1)$$

EXPERIMENTAL LAB: FIELD APPLICATIONS

Problem 1: A surveyor places a laser at origin $(0,0)$ and a reflector at $(15, 20)$. Calculate the distance between them.

Chapter 2

Coordinate Geometry

2.1 Introduction

Coordinate geometry, also known as analytic geometry, is the study of geometry using a coordinate system. This contrasts with synthetic geometry.

2.2 The Cartesian Plane

The Cartesian plane is defined by two perpendicular number lines: the x-axis, which is horizontal, and the y-axis, which is vertical.